SENG 4620
Practical Cloud Computing

Midterm EXAM - Sample

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Exam duration: 80 minutes

Question	Score	Points
01		20
02		20
03		16
04		20
05		24
Total		100

This is a CLOSED BOOK exam. Textbooks, notes, laptops, personal digital assistants, tablets, and cellular phones are NOT allowed.

It is a 180 minute exam, with a total of 100 marks. There are 5 sections of questions, and 14 pages (including this cover page). Please read each question carefully, and write your answers legibly in the space provided. You may do the questions in any order you wish, but please USE YOUR TIME WISELY.

When you are finished, please hand in your exam paper and sign out.
Good luck!

Student Name: _____ Score: _____ / 100

Student ID: _____

Q1 - Secure Access

1. According to the AWS Shared Responsibility Model, AWS is responsible for managing IAM users, groups, and policies on behalf of the customer. (1 Mark)

(a) True

(b) False

Answer: (b) False. The customer is responsible for security "in" the cloud, which includes managing identity and access management (IAM) permissions for their own users, groups, and roles.

2. When writing an identity-based IAM policy to attach to a user, the "Principal" element must be included to specify which user the policy applies to. (1 Mark)

(a) True

(b) False

Answer: (b) False. For an identity-based policy, the principal is *implied* as the user, group, or role that the policy is attached to. The "Principal" element is only required in resource-based policies (or role trust policies).

3. What types of credentials are required for an IAM user to sign in to the AWS Management Console and to use the AWS Command Line Interface (CLI), respectively? (2 Marks)

(a) Console: Username and Password. CLI: Access Key ID and Secret Access Key.

(b) Console: Access Key ID and Secret Access Key. CLI: Username and Password.

(c) Console: Username and Password. CLI: Username and Password.

(d) Console: Access Key ID. CLI: Secret Access Key only.

Answer: a: Access to the web-based Management Console requires a Username and Password. Programmatic access via the CLI or SDKs requires an Access Key ID and Secret Access Key.

4. A key difference between an IAM User and an IAM Role is that an IAM User is associated with _____ credentials, while an IAM Role is associated with _____ credentials. (2 Marks)

(a) temporary; long-term

(b) long-term; temporary

(c) shared; individual

(d) encrypted; unencrypted

Answer: b: An IAM User has long-term credentials (a password or access keys) that do not expire. An IAM Role provides temporary security credentials that are automatically rotated and expire after a short duration.

5. What is the fundamental difference between an **identity-based policy** and a **resource-based policy** in IAM? (4 Marks)

Answer: The fundamental difference is what they are attached to:

- An **identity-based policy** is attached to an IAM identity (a user, group, or role). It answers the question, "What can this identity do?"
- A **resource-based policy** is attached directly to an AWS resource (e.g., an S3 bucket, an SQS queue). It answers the question, "Who is allowed to access this resource?"

6. What is an "AWS access key"? Explain (1) its purpose and (2) why it is considered a long-term credential that poses a security risk if not managed properly. (4 Marks)

Answer:

1. **Purpose:** An AWS access key (composed of an Access Key ID and a Secret Access Key) is a credential used for programmatic access to AWS services. It allows applications, the AWS CLI, or scripts to make authenticated requests.
2. **Security Risk:** It is a "long-term" credential because it never expires on its own. If it is leaked or compromised, an attacker can use it to access your account until you manually revoke it. This is why they should be rotated regularly or, preferably, replaced by using temporary credentials via IAM roles.

7. A company (Account A) wants to grant a data science team (IAM User 'DataScientist') in a different AWS account (Account B) read-only access to an S3 bucket (named 'research-data') located in Account A. Describe the high-level steps required in both accounts to set this cross-account access up securely using an IAM Role. (8 Marks)

Answer: This requires a two-part setup involving an IAM Role in the trusting account (A) and policy in the trusted account (B).

1. **In Account A (The Trusting Account - where the bucket is):**
   - **Create an IAM Role:** Create a new IAM Role.
   - **Define Trust Policy:** In the role's Trust Policy, specify Account B (e.g., `"Principal": {"AWS": "arn:aws:iam::ACCOUNT-B-ID:root"}`) as the trusted principal. This allows Account B to assume this role.
   - **Define Permissions Policy:** Attach a permissions policy to the role that grants read-only access (e.g., `s3:GetObject`) to the 'research-data' bucket (e.g., `"Resource": "arn:aws:s3:::research-data/*"`).
2. **In Account B (The Trusted Account - where the user is):**
   - **Create an IAM Policy:** Create a new policy that grants the 'DataScientist' user permission to call `sts:AssumeRole` on the specific ARN of the role created in Account A.
   - **Attach Policy:** Attach this new policy to the 'DataScientist' IAM User (or a group they are in).
3. **Access:** The 'DataScientist' user can now programmatically assume the role in Account A. This provides them with temporary credentials that have the read-only permissions defined by the role.

8. You are an administrator for a company that wants to increase security. You have a policy that allows your 'Admin' IAM group to terminate EC2 instances. However, you must restrict this powerful action so that it can only be performed from the corporate office, which has a public IP address range of `198.51.100.0/24`. **Task:** Write a complete JSON IAM policy statement that **denies** the `ec2:TerminateInstances` action for all resources **UNLESS** the request comes from the specified IP range. (10 Marks)

1 1. Amazon S3 is primarily a File Storage service, designed to store data in a hierarchical directory structure similar to a network drive. (1 Mark)

(a) True

(b) False

Answer: (b) False. Amazon S3 is an Object Storage service. Data is stored as objects with keys and metadata in a flat structure, not as files in a hierarchical structure.

1 2. Data transferred out from an Amazon S3 bucket to the public internet is always free of charge, regardless of the amount. (1 Mark)

(a) True

(b) False

Answer: (b) False. While data transfer *in* from the internet is free, data transfer *out* to the internet is only free for the first 100 GB per month; charges apply beyond that amount.

2 3. A company needs to archive large volumes of data for 7-10 years to meet compliance requirements. The data will be accessed extremely rarely (perhaps once or twice a year), and retrieval times of 12 hours or more are acceptable. Which S3 storage class offers the lowest storage cost for this scenario? (2 Marks)

(a) S3 Standard-IA

(b) S3 Glacier Flexible Retrieval

(c) S3 Intelligent-Tiering

(d) S3 Glacier Deep Archive

Answer: d: S3 Glacier Deep Archive is the lowest-cost storage class, specifically designed for long-term retention (7-10+ years) and digital preservation of data accessed rarely, meeting regulatory requirements.

2 4. What is the primary function of enabling Versioning on an Amazon S3 bucket? (2 Marks)

(a) To automatically encrypt objects as they are uploaded.

(b) To reduce storage costs by compressing older objects.

(c) To protect objects from accidental overwrites and deletions.

(d) To speed up object retrieval by caching multiple copies.

Answer: c: Amazon S3 versioning protects objects from accidental overwrites and deletes by preserving previous versions when an object is updated or a delete marker when it's deleted.

4 5. Amazon S3 has a flat namespace for objects within a bucket. How can users organize objects to simulate a hierarchical folder structure, and how does this affect object keys? (4 Marks)

Answer: S3 simulates folders by using shared **name prefixes** for objects. Users include characters like '/' in the object key name to denote structure (e.g., `images/2024/photo.jpg`). The full string, including the prefixes and the filename (e.g., `images/2024/photo.jpg`), becomes the object's unique **key**. Prefixes allow filtering, mimicking folder navigation.

4 6. Define S3 **Durability** and S3 **Availability**. What are the design targets for S3 Standard for each of these metrics? (4 Marks)

Answer:

- **Durability:** Refers to the protection against the loss of an object. S3 Standard is designed for **99.999999999% (11 nines)** durability.
- **Availability:** Refers to the ability to access and retrieve your data when needed. S3 Standard is designed for **99.99% (4 nines)** availability.

8 7. Compare and contrast the different approaches to encrypting data at rest in Amazon S3: Server-Side Encryption with S3-Managed Keys (SSE-S3), Server-Side Encryption with KMS-Managed Keys (SSE-KMS), and Client-Side Encryption (CSE). Discuss who manages the encryption keys and the process in each case. (8 Marks)

Answer: All three methods protect data at rest, but differ in key management and process control:

- **SSE-S3 (Server-Side Encryption with S3-Managed Keys):**
  - **Process:** Amazon S3 automatically encrypts objects before saving them to disk and decrypts them upon download. This is the default encryption setting for all new buckets.
  - **Key Management:** Amazon S3 manages the encryption keys. Users have minimal control but benefit from simplicity and no additional cost beyond standard S3 charges.
- **SSE-KMS (Server-Side Encryption with AWS KMS-Managed Keys):**
  - **Process:** Similar to SSE-S3, S3 handles encryption/decryption, but it uses keys managed within AWS Key Management Service (KMS).
  - **Key Management:** AWS KMS manages the keys, but the user has more control. Users can create, rotate, and manage Customer Managed Keys (CMKs) or use AWS Managed Keys. This provides an audit trail (via CloudTrail) showing when and by whom keys were used. There are additional costs associated with KMS usage.
- **CSE (Client-Side Encryption):**
  - **Process:** The user encrypts the data *before* uploading it to S3. S3 stores the already-encrypted data. The user must decrypt the data after downloading it.
  - **Key Management:** The user manages the entire encryption process, including generating, securing, and rotating the encryption keys. This offers the maximum control but also the most operational overhead.

10 8. A company generates daily transaction reports. These reports are frequently accessed for the first week (7 days) for verification. After that, they are accessed infrequently for up to 90 days for monthly summaries. After 90 days, they must be kept for 5 years for compliance but are almost never accessed. After 5 years, they should be deleted. **Task:** Design a cost-effective storage strategy using S3 storage classes and a Lifecycle policy to automate this process. Specify the initial storage class, the transition rules (storage class and timing), and the final expiration rule. Justify your choices. (10 Marks)

Answer: This requires a multi-stage lifecycle policy to move data to progressively cheaper storage tiers.

1. **Initial Storage (Day 0):** Upload reports to **S3 Standard**. *Justification:* Provides low latency and high throughput needed for frequent access during the first 7 days.
2. **Lifecycle Rule 1 - Transition (Day 7):** Transition objects to **S3 Standard-IA** (Infrequent Access) after 7 days. *Justification:* Offers lower storage cost than S3 Standard for data accessed infrequently, suitable for the period up to 90 days. Retrieval fees apply but are acceptable for monthly summaries.
3. **Lifecycle Rule 2 - Transition (Day 90):** Transition objects from S3 Standard-IA to **S3 Glacier Deep Archive** after 90 days. *Justification:* This is the lowest-cost S3 storage class, ideal for long-term archival (5 years) where data is rarely accessed, and long retrieval times (hours) are acceptable for compliance purposes.
4. **Lifecycle Rule 3 - Expiration (Day 1825):** Set an expiration action to permanently delete objects after 1825 days (approx. 5 years). *Justification:* Automates the deletion of data that is no longer required for compliance, eliminating storage costs.

10 9. Your company uses an S3 bucket to migrate a 10 TB dataset, consisting of millions of small image files and several very large video files (up to 100 GB each), from your on-premises data center to an Amazon S3 bucket located in a different country. The migration needs to be completed as quickly as possible over your existing internet connection. **Task:** Describe a strategy for performing this migration efficiently and reliably. Include recommendations for (1) handling the large video files, (2) accelerating the overall transfer speed across the long distance, and (3) the general method for initiating the transfers. (10 Marks)

Answer: A strategy combining multiple S3 features and tools is recommended for efficiency and reliability.

1. **Handling Large Video Files (Multipart Upload):** For the very large video files (over 100 MB, definitely for 100 GB files), ensure the transfer tool uses **S3 Multipart Upload**. *Benefits:* This breaks large files into smaller parts that can be uploaded in parallel, improving throughput. It also allows for quicker recovery if a part fails (only the failed part needs re-uploading) and allows pausing/resuming. Most AWS tools (CLI, SDKs) handle this automatically for large files.

2. **Accelerating Transfer Speed (S3 Transfer Acceleration):** Enable **S3 Transfer Acceleration** on the destination S3 bucket. *Benefit:* This feature uses Amazon CloudFront's globally distributed edge locations to optimize the network path for data traveling long distances over the public internet. Data is ingested at a nearby edge location and travels over the AWS backbone to the S3 bucket, which can significantly improve speed (50-500

3. **Initiating Transfers (AWS CLI/SDK or AWS DataSync):**

   - **AWS CLI/SDK:** Use the AWS Command Line Interface (`aws s3 sync` or `aws s3 cp`) or scripts using AWS SDKs. These tools support multipart upload and can be configured to use the Transfer Acceleration endpoint. This requires scripting or manual execution but offers flexibility.
   - **AWS DataSync (Alternative/Recommended):** For a large-scale migration like this, consider AWS DataSync. It's a managed service designed for large data transfers between on-premises storage and AWS storage services like S3. It automates many tasks, handles network optimization, performs data integrity checks, and integrates well with S3 features. While not explicitly covered in detail in the slides, it's a relevant AWS service for this scale. (Note: Transfer Family is more for protocol-based transfers like SFTP/FTP).

Using Transfer Acceleration combined with Multipart Upload (handled by the chosen tool like CLI/SDK or DataSync) provides the fastest and most reliable method over the existing internet connection.

Q3 - Compute

1 1. An EC2 instance is a physical server located in an AWS data center that you have exclusive access to. (1 Mark)

(a) True

(b) False

Answer: (b) False. An EC2 instance is a Virtual Machine (VM) that runs on a physical host server managed by AWS. Multiple instances typically share the same physical hardware, managed by a hypervisor. (Dedicated Hosts/Instances are an exception, but not the default).

1 2. The price for EC2 Spot Instances is fixed for the duration of the instance runtime, similar to On-Demand instances. (1 Mark)

(a) True

(b) False

Answer: (b) False. Spot Instance prices fluctuate based on long-term supply and demand for spare EC2 capacity. Your instance runs as long as capacity is available and your bid price (or the default maximum, the On-Demand price) exceeds the current Spot price.

2 3. You have multiple Linux-based EC2 instances that need to concurrently access and modify data on a shared, scalable file system. Which AWS storage service is designed for this use case? (2 Marks)

(a) Amazon S3

(b) Amazon EBS Multi-Attach

(c) Amazon EFS

(d) Instance Store

Answer: c: Amazon Elastic File System (EFS) provides a fully managed, scalable file system for Linux-based workloads, supporting NFS protocols and allowing concurrent access from multiple EC2 instances. EBS Multi-Attach is for specific instance types/volumes, S3 is object storage, and Instance Store is local/ephemeral.

2 4. What is the purpose of EC2 instance metadata, and how can a running instance access it? (2 Marks)

(a) It's user-provided data for instance configuration, accessed via the EC2 console.

(b) It's billing information about the instance, accessed via the AWS Billing console.

(c) It's information about the instance (like its ID, IP address, AMI ID), accessible via a special HTTP endpoint from within the instance.

(d) It's performance metrics (like CPU utilization), accessed via Amazon CloudWatch.

Answer: c: Instance metadata is data about the instance itself (e.g., instance ID, IP addresses, hostname, security groups). It can be accessed from within the running instance by making an HTTP request to a special link-local address: http://169.254.169.254/latest/meta-data/.

4 5. What is the primary performance difference between SSD-backed EBS volumes (like gp2, io1) and HDD-backed EBS volumes (like st1, sc1)? (4 Marks)

Answer: SSD-backed volumes are optimized for transactional workloads where the dominant performance attribute is **IOPS** (Input/Output Operations Per Second). They are good for small, frequent read/write operations like databases. HDD-backed volumes are optimized for large, streaming workloads where the dominant performance attribute is **Throughput** (measured in MiB/s). They are good for big data, data warehouses, and log processing.

4 6. Explain the concept of an EC2 Placement Group using the "Cluster" strategy. What is the main benefit of using this strategy? (4 Marks)

Answer: A Cluster Placement Group is a logical grouping of interdependent instances packed closely together **within a single Availability Zone**. The main benefit is achieving **low-latency network performance** between the instances in the group. This is ideal for applications that require high levels of inter-node communication, such as high-performance computing (HPC) applications.

8 7. Describe the three main AMI deployment models: Basic AMI, Silver AMI, and Golden AMI. For each model, explain (1) how much configuration is "baked into" the AMI, (2) the typical impact on instance boot time vs. AMI build time, and (3) a suitable use case. (8 Marks)

Answer: The three AMI deployment models represent different strategies for pre-configuring instances:

1. **Basic AMI:**
    - **Configuration:** Contains only the base Operating System (OS). All application installations and configurations must be done after launch, typically via User Data scripts or configuration management tools.
    - **Time Impact:** Very short AMI build time, but much slower instance boot/initialization time because all setup happens at launch.
    - **Use Case:** Starting point for creating other AMIs, environments where flexibility and latest versions at launch are prioritized over boot speed.
2. **Silver AMI (Mutable Model):**
    - **Configuration:** Contains the OS plus common prerequisite software, utilities, or agents ("half-baked"). Some application-specific configuration still needs to be done after launch (manually or via User Data).
    - **Time Impact:** Moderate AMI build time, moderate instance boot time. Provides a balance between the two extremes.
    - **Use Case:** Environments where a common baseline is useful across different instance roles, but specific configurations vary. Offers easier rollbacks than Golden AMIs. AWS Managed Services (AMS) provides AMIs for this model.
3. **Golden AMI (Immutable Model):**
    - **Configuration:** Contains the OS plus all application code, dependencies, and configurations fully "baked in." Instances launched from this AMI are ready to serve traffic almost immediately. Updates require creating a completely new Golden AMI.
    - **Time Impact:** Longest AMI build time, but shortest instance boot time.
    - **Use Case:** Environments prioritizing fast boot times and consistency, auto-scaling groups where new instances must come online quickly, immutable infrastructure deployments.

8. A company wants to automate the deployment of its web application onto new EC2 instances. When a new Linux EC2 instance is launched, it should automatically:

- Install the necessary web server software (e.g., Apache).
- Download the latest version of the web application code from a private S3 bucket.
- Start the web server application.

Task: Explain how you would use **EC2 User Data** and potentially **Instance Metadata** (especially considering access to a private S3 bucket) to achieve this automated configuration without manual intervention after launch. Describe the role each component plays in the process. (10 Marks)

Answer: This automation can be achieved by combining EC2 User Data with an IAM Role accessed via Instance Metadata.

1. **Role of EC2 User Data:** User Data is the primary mechanism for executing commands upon the first boot of an EC2 instance. A script would be provided in the User Data field during instance launch. This script would contain the commands to:
    - Use the OS package manager (like `yum` or `apt`) to install the Apache web server software.
    - Use the AWS CLI or SDK (which needs to be installed or available on the AMI) to copy the application code from the specified private S3 bucket to the appropriate directory on the instance (e.g., `/var/www/html`).
    - Use system commands (like `systemctl start httpd`) to start the web server service.
2. **Role of Instance Metadata (and IAM Roles):** To securely access the *private* S3 bucket without embedding credentials in the User Data script (which is insecure), an **IAM Role** must be used:
    - An IAM Role with permissions to read (`s3:GetObject`) from the specific S3 bucket is created.
    - This IAM Role is attached to the EC2 instance via an Instance Profile.
    - The AWS CLI/SDK running within the User Data script will automatically retrieve temporary security credentials associated with this role from the **Instance Metadata Service** (available at `http://169.254.169.254/...`) [cite: 2979-2983].
    - These temporary credentials are then used to authenticate the request to S3, allowing the script to securely download the application code without needing hardcoded keys.

By combining User Data (to execute the steps) and an IAM Role accessed via Instance Metadata (to provide secure S3 access), the entire configuration process is automated securely.

10 9. You are designing the storage configuration for an EC2 instance that will host a critical relational database (e.g., PostgreSQL, MySQL). The database requires persistent storage and must handle thousands of transactions per second, demanding very high, consistent IOPS performance with low latency. Cost is a secondary concern to performance and reliability. **Task:** Recommend an appropriate Amazon EBS volume type for the database data files. Explain why this type is suitable and mention an instance-level configuration that ensures optimal performance between the instance and the EBS volume. (10 Marks)

Answer:

1. **Recommended EBS Volume Type: Provisioned IOPS SSD (io1 or io2/io2 Block Express)**.

2. **Justification:**

   - **High Performance (IOPS):** Provisioned IOPS SSD volumes are the highest-performance EBS SSD volumes, specifically designed for I/O-intensive workloads like critical databases that require sustained and consistent IOPS performance. They allow you to specify the exact IOPS rate you need (up to a high limit), ensuring predictable performance.
   - **Low Latency:** SSD technology inherently provides lower latency compared to HDD, which is crucial for transactional database responsiveness.
   - **Persistence:** Like all EBS volumes, io1/io2 provide persistent block storage, ensuring data survives instance stops and starts.
   - **Reliability:** Designed for mission-critical applications. io2/io2 Block Express offer even higher durability than io1/gp3/gp2.

   (General Purpose SSD (gp3/gp2) might be considered but may not guarantee the *consistent* high IOPS needed for sustained peak loads as effectively as Provisioned IOPS volumes do).

3. **Instance-Level Configuration for Optimal Performance:** Ensure the chosen EC2 instance type supports **EBS Optimization**. EBS-optimized instances provide dedicated network bandwidth between the EC2 instance and its attached EBS volumes, minimizing contention between EBS I/O and other network traffic from the instance. For instance types built on the AWS Nitro System, EBS optimization offers even higher performance. For many modern instance types, EBS optimization is enabled by default, but for others, it might need to be explicitly enabled at launch.

Q4 - Database Layer

1 1. Amazon Aurora is compatible only with MySQL and does not offer a PostgreSQL-compatible edition. (1 Mark)

(a) True

(b) False

Answer: (b) False. Amazon Aurora is a relational database management system built for the cloud with full MySQL and PostgreSQL compatibility.

1 2. AWS Database Migration Service (AWS DMS) can only be used for homogeneous migrations (e.g., MySQL to MySQL), not for heterogeneous migrations (e.g., Oracle to PostgreSQL). (1 Mark)

(a) True

(b) False

Answer: (b) False. AWS DMS supports both homogeneous migrations (same engine source and target) and heterogeneous migrations (different engine source and target). Heterogeneous migrations often require schema conversion using AWS SCT or AWS DMS Schema Conversion.

2 3. Which AWS service is primarily used to provide an in-memory cache to improve the read performance of applications by reducing latency, often used in front of databases like RDS or DynamoDB? (2 Marks)

(a) Amazon Redshift

(b) Amazon Neptune

(c) Amazon ElastiCache

(d) Amazon DocumentDB

Answer: c: Amazon ElastiCache is the AWS service designed for deploying, operating, and scaling in-memory data stores or caches in the cloud, commonly used to improve application read performance.

2 4. A company wants to migrate its on-premises Oracle database to Amazon Aurora PostgreSQL. Since the source and target database engines are different, which combination of AWS services is typically recommended for this heterogeneous migration? (2 Marks)

(a) AWS DMS only

(b) AWS SCT or DMS Schema Conversion only

(c) Amazon RDS and Amazon EC2

(d) AWS DMS and AWS SCT (or DMS Schema Conversion)

Answer: d: For heterogeneous migrations, AWS DMS is used to migrate the data, while the AWS Schema Conversion Tool (AWS SCT) or the newer AWS DMS Schema Conversion is used to convert the source schema and code (e.g., stored procedures, functions) to be compatible with the target database engine.

4 5. What is the primary use case for Amazon Redshift, and how does its focus differ from transactional databases like Amazon RDS? (4 Marks)

Answer: Primary Use Case: Amazon Redshift is a fully managed, cloud-based **data warehousing** service designed for **Online Analytical Processing (OLAP)** and handling petabyte-scale analytics workloads. Difference from RDS: While RDS is optimized for **Online Transaction Processing (OLTP)** - handling frequent reads and writes for application transactions - Redshift is optimized for complex analytical queries across large datasets, often using columnar storage for better query performance.

4 6. Explain the concept of Amazon DynamoDB Global Tables and state their main benefit for applications with a global user base. (4 Marks)

Answer: Concept: DynamoDB Global Tables provide a fully managed solution for deploying a **multi-Region, multi-active database**. You select the AWS Regions where you want the table to be available, and DynamoDB automatically propagates ongoing data changes between the replica tables in those Regions. Main Benefit: They enable applications to deliver **fast, low-latency data access** to users regardless of their geographic location by allowing them to read and write data locally to the nearest AWS Region. They also provide increased fault tolerance.

8 7. Compare the use of Amazon Aurora Serverless with provisioned Amazon Aurora instances. Discuss the key differences in terms of (1) capacity management, (2) scaling behavior, (3) performance characteristics, and (4) cost model, mentioning suitable use cases for each. (8 Marks)

Answer: Aurora Serverless and provisioned Aurora instances offer different approaches to managing database capacity and cost:

1. **Capacity Management:**
   - **Provisioned Aurora:** Requires manual capacity management. You select an instance size (CPU/RAM) and must monitor and manually adjust it if the workload changes.
   - **Aurora Serverless:** Provides hands-off, automatic capacity management. The database automatically starts up, shuts down, and scales capacity (compute and memory) based on application demand within a configured range.
2. **Scaling Behavior:**
   - **Provisioned Aurora:** Scaling typically involves modifying the instance class, which can cause a brief downtime (though Multi-AZ minimizes this for failover). Read scaling is done by adding read replicas.
   - **Aurora Serverless:** Offers fine-grained, automatic scaling. It can scale compute capacity up or down rapidly, sometimes in seconds, often without disrupting connections (especially Serverless v2). It can also scale down to zero when idle (Serverless v1).
3. **Performance Characteristics:**
   - **Provisioned Aurora:** Delivers consistent, predictable performance based on the selected instance size. Ideal for steady-state workloads with known performance requirements.
   - **Aurora Serverless:** Performance scales with demand. Serverless v1 could experience brief pauses during scaling or cold starts. Serverless v2 offers more granular scaling with less performance impact. May not be ideal for sustained high-performance needs compared to a large provisioned instance.
4. **Cost Model Use Cases:**
   - **Provisioned Aurora:** Billed per instance-hour based on the chosen instance size, regardless of actual usage. Suitable for predictable, steady-state workloads where utilization is consistently high.
   - **Aurora Serverless:** Billed based on Aurora Capacity Units (ACUs) consumed per second. Can be more cost-effective for variable, intermittent, or unpredictable workloads, new applications with unknown load, development/testing environments, or applications with infrequent usage, as it can scale down (even to zero in v1) when idle.

10 8. You are designing the backend for a social networking application. Key requirements include storing user profiles, user posts, and the complex relationships between users (e.g., who follows whom, friendships, group memberships). The application needs to perform rapid queries to find connections, like "find all friends of my friends who live in Vancouver" or "find all users who liked a specific post." **Task:** Recommend a suitable AWS purpose-built database service for managing the relationship data (users and their connections). Justify why this service is a better fit for this specific requirement than a traditional relational database (like Amazon RDS). (10 Marks)

Answer:

1. **Recommended Service: Amazon Neptune**.
2. **Justification:**
    - **Data Model Fit:** Neptune is a purpose-built **graph database**. Social networks are inherently graph-like structures, consisting of entities (users, posts) and the relationships (follows, likes, friend-of) connecting them. Neptune stores these relationships as first-class citizens, making it natural and efficient to model and query social connections.
    - **Query Performance for Relationships:** Graph databases like Neptune are optimized for traversing relationships. Queries like "friends of friends" or finding common connections are typically much faster and simpler to express in graph query languages (like Gremlin or SPARQL, supported by Neptune) compared to recursive SQL JOINs in a relational database. Complex relationship queries in RDS can become slow and resource-intensive as the network grows.
    - **Scalability for Connected Data:** Neptune is designed to handle large, highly connected datasets efficiently, which is characteristic of social networks. While RDS can scale, managing and querying deep relationships across potentially billions of connections can become a bottleneck.
    - **Flexibility:** While Neptune has a schema, graph models can often be more flexible in adding new types of entities or relationships compared to altering rigid table structures in RDS.

    While user profiles or posts *could* be stored in RDS or DynamoDB, managing the complex and frequently queried *relationships* between users is where Neptune provides a significant advantage over a traditional relational database.

12 9. A company needs to migrate a 500 GB on-premises MySQL database to Amazon RDS for MySQL. They want to minimize the downtime required for the final cutover. The network connection between the on-premises data center and AWS is reliable but not extremely fast. **Task:** Outline the high-level steps for performing this migration using AWS Database Migration Service (AWS DMS) with Change Data Capture (CDC). Explain the role of (1) the replication instance, (2) source and target endpoints, (3) the full load task, and (4) the CDC task in achieving a minimal downtime migration. (12 Marks)

Answer: Using AWS DMS with CDC allows for minimal downtime migration by performing an initial bulk load and then replicating ongoing changes.

1. **Setup Replication Instance:** Provision an **AWS DMS Replication Instance** in AWS. This is an EC2 instance managed by DMS that performs the actual data transfer between the source and target. Size it appropriately based on the database size and change volume.

2. **Define Endpoints:**

   - Create a **Source Endpoint** in DMS that points to the on-premises MySQL database. This requires providing connection details (hostname, port, credentials) and ensuring network connectivity (e.g., via VPN or Direct Connect) between the replication instance and the on-premises database.
   - Create a **Target Endpoint** in DMS that points to the newly created (and initially empty) Amazon RDS for MySQL instance. Provide connection details for the RDS instance.

3. **Create Full Load + CDC Replication Task:** Create a DMS Replication Task configured for "Migrate existing data and replicate ongoing changes" (Full Load + CDC).

   - **Full Load Phase:** When started, the task first performs a **full load**, copying the existing 500 GB of data from the source endpoint (on-premises MySQL) to the target endpoint (RDS MySQL) via the replication instance. During this phase, changes occurring on the source are cached by DMS.
   - **Change Data Capture (CDC) Phase:** After the full load completes, DMS switches to the **CDC phase**. It applies the cached changes and then continues to capture ongoing changes (INSERTs, UPDATEs, DELETEs) from the source database's binary logs (binlogs) and replicates them to the target RDS instance in near real-time. This keeps the source and target databases synchronized.

4. **Monitor Replication Lag:** Monitor the DMS task to ensure replication lag (the delay between a change on the source and its appearance on the target) is minimal.

5. **Cutover:** Once the target is fully synchronized and replication lag is near zero, schedule a brief maintenance window. Stop application writes to the on-premises database, wait for DMS to apply any final changes to RDS, validate the data on RDS, and then re-point the application connection strings to the Amazon RDS endpoint. This cutover window represents the minimal downtime required.

6. **Cleanup:** After successful cutover and testing, stop the DMS task and decommission the on-premises database and DMS resources.

Q5 - Networks

1 1. By default, a newly created Security Group allows all inbound traffic but denies all outbound traffic. (1 Mark)

(a) True

(b) False

Answer: (b) False. By default, a new Security Group denies all inbound traffic and allows all outbound traffic.

1 2. VPC Flow Logs capture the full packet content of the IP traffic going to and from network interfaces in your VPC. (1 Mark)

(a) True

(b) False

Answer: (b) False. VPC Flow Logs capture metadata about the IP traffic (like source/destination IPs, ports, protocol, packet/byte counts, action), not the actual content of the packets. Traffic Mirroring is used to capture packet content.

2 3. A Network Access Control List (Network ACL) operates at which level within an Amazon VPC? (2 Marks)

(a) The individual EC2 instance level.

(b) The Availability Zone level.

(c) The VPC level.

(d) The subnet level.

Answer: d: A Network ACL acts as an optional firewall for controlling traffic in and out of one or more subnets. It operates at the subnet level, unlike Security Groups which operate at the instance/ENI level.

2 4. What is the primary purpose of allocating an Elastic IP address and associating it with an EC2 instance in a public subnet? (2 Marks)

(a) To provide a private IP address for internal communication.

(b) To automatically scale the instance based on traffic.

(c) To provide a static, public IPv4 address that persists even if the instance is stopped or terminated.

(d) To encrypt traffic between the instance and the internet gateway.

Answer: c: An Elastic IP address is a static, public IPv4 address designed for dynamic cloud computing. Unlike default public IPs, it can be associated with your account and re-associated with a different instance if the original one fails or is stopped, providing a persistent public endpoint.

4 5. What is the role of an Internet Gateway in an Amazon VPC? (4 Marks)

Answer: An Internet Gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between resources in your VPC (specifically, those in public subnets with a route to it) and the internet. It serves two main purposes: (1) provides a target in VPC route tables for internet-routable traffic, and (2) performs Network Address Translation (NAT) for instances that have been assigned public IPv4 addresses.

4 6. Briefly describe the difference between an Interface VPC Endpoint and a Gateway VPC Endpoint in terms of (1) underlying technology and (2) supported AWS services. (4 Marks)

Answer:

1. **Underlying Technology:** Interface VPC Endpoints use **AWS PrivateLink** technology, creating an Elastic Network Interface (ENI) with a private IP address within your subnet to act as the entry point to the service. Gateway VPC Endpoints use **route table entries**; you add a route to your subnet's route table targeting the endpoint for a specific service's public IP range (using prefix lists).
2. **Supported Services:** Interface VPC Endpoints (AWS PrivateLink) support a wide range of AWS services, partner services, and custom services. Gateway VPC Endpoints support only **Amazon S3** and **Amazon DynamoDB**.

8 7. Compare and contrast Security Groups and Network ACLs in Amazon VPC. Discuss their differences based on the following four criteria: (1) Scope of Operation, (2) Rule Types Supported, (3) Statefulness, and (4) Default Rules for newly created resources. (8 Marks)

Answer: Security Groups and Network ACLs are both firewall mechanisms but operate differently:

1. **Scope of Operation:**
   - **Security Groups:** Operate at the resource level, specifically associated with an Elastic Network Interface (ENI) of an EC2 instance or other AWS resource (like RDS).
   - **Network ACLs:** Operate at the subnet level, controlling traffic entering or leaving an entire subnet.
2. **Rule Types Supported:**
   - **Security Groups:** Only support **Allow** rules. Traffic that doesn't match an allow rule is implicitly denied.
   - **Network ACLs:** Support both **Allow** and **Deny** rules explicitly.
3. **Statefulness:**
   - **Security Groups:** Are **stateful**. If outbound traffic is allowed (or initiated based on an inbound allow rule), the return traffic is automatically allowed back in, regardless of inbound rules.
   - **Network ACLs:** Are **stateless**. Return traffic must be explicitly allowed by a corresponding inbound or outbound rule.
4. **Default Rules (for new resources):**
   - **Security Groups:** A newly created security group denies all inbound traffic and allows all outbound traffic.
   - **Network ACLs:** The default Network ACL created with a VPC allows all inbound and all outbound traffic. A newly *custom* created Network ACL denies all inbound and all outbound traffic until rules are added.

10 8. You suspect intermittent connectivity issues between two EC2 instances within the same VPC, possibly due to overly restrictive security group or network ACL rules. You need to capture detailed information about the IP traffic flow to diagnose the problem. **Task:**

Design a monitoring solution using VPC Flow Logs. Specify (1) the possible scopes where you can enable Flow Logs, (2) the two primary destinations where Flow Logs can publish data, and (3) explain how analyzing the 'action' and 'log-status' fields in the flow log records could help identify dropped packets or configuration errors. (10 Marks)

Answer:

1. **Possible Scopes for Enabling Flow Logs:** You can enable VPC Flow Logs at three different levels to capture IP traffic metadata:

   - The entire **VPC**.
   - A specific **subnet**.
   - A specific **Elastic Network Interface (ENI)** (attached to an EC2 instance, for example).

   For targeted troubleshooting between two instances, enabling flow logs on their respective ENIs might be most efficient.

2. **Primary Destinations for Flow Logs:** Flow log data can be published to:

   - **Amazon CloudWatch Logs:** Allows near real-time viewing, searching, and alarming based on log data via the AWS Console or API.
   - **Amazon S3:** Stores log files in an S3 bucket, suitable for archival, batch processing, or querying using tools like Amazon Athena.

3. **Analyzing 'action' and 'log-status' fields:**

   - **'action' field:** This field indicates whether the traffic recorded was **ACCEPTED** or **REJECTED**. If you see traffic between the two instances being consistently REJECTED, it strongly suggests that either a Security Group rule or a Network ACL rule is blocking the traffic. Examining the source/destination IPs, ports, and protocols in the rejected record can help pinpoint which rule is responsible.
   - **'log-status' field:** This field indicates the status of the flow logging process itself. While "OK" is normal, "NODATA" simply means no traffic was seen in the interval. However, seeing "**SKIPDATA**" indicates that some flow log records were missed during the interval due to capacity constraints or internal errors. While less common for direct connectivity issues, seeing SKIPDATA during periods of high traffic might indicate the monitoring itself is missing information. Primarily, the 'action' field is key for diagnosing blocked traffic.

10 9. An application running on an EC2 instance in a private subnet (`10.0.100.0/24`) needs to frequently access the Amazon S3 API to read and write objects in a specific bucket (`my-app-data-bucket`). Company policy prohibits routing any traffic destined for AWS services over the public internet for security reasons. **Task:** Design a secure and efficient solution using a VPC Endpoint to allow this private communication. Specify (1) the type of VPC Endpoint required for S3 that avoids using AWS PrivateLink, (2) how this endpoint is configured in relation to the subnet's route table, and (3) the benefit of using this endpoint compared to routing through a NAT Gateway. (10 Marks)

Answer:

1. **Type of VPC Endpoint:** A **Gateway VPC Endpoint** for Amazon S3 is required. This type of endpoint provides private connectivity specifically to S3 (and DynamoDB) without using AWS PrivateLink or requiring network interfaces with private IPs within the subnet.

2. **Route Table Configuration:** When creating the Gateway VPC Endpoint for S3, you must specify the **route table(s)** associated with the private subnet(s) that need access (in this case, the route table for `10.0.100.0/24`). AWS automatically adds a specific route to this route table. The route uses a *prefix list* provided by AWS (representing S3's public IP range in the Region) as the **Destination**, and the Gateway VPC Endpoint ID as the **Target**. This ensures that traffic from the private subnet destined for the S3 service is routed directly through the endpoint over the AWS backbone, rather than going to an Internet Gateway or NAT Gateway.

3. **Benefit Compared to NAT Gateway:**

   - **Security:** Traffic does not traverse the public internet or pass through a NAT device, keeping it entirely within the AWS network. This enhances security and meets the company policy requirement.
   - **Cost:** Gateway VPC Endpoints for S3 do **not** incur hourly charges or data processing fees, unlike NAT Gateways or Interface VPC Endpoints. NAT Gateways charge for data processing and per hour.
   - **Performance/Bandwidth:** Gateway VPC Endpoints do not have the bandwidth constraints associated with NAT Gateways or the initial limits of Interface Endpoints.

*** THE END ***